

Notes: Cagé, Hervé & Viaud (RES, 2020)

The Production of Information in an Online World

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1 Big picture

- **Question.** How much online news is original, how quickly do media outlets reproduce competitors' stories, and do original producers receive enough audience to preserve incentives for newsgathering?
- **Setting.** The authors study 86 major French online media outlets in 2013, including newspapers, television, radio, pure online media, AFP, and Reuters.
- **Core fact.** Online media are extremely reactive: average reaction time is 169 minutes, the median is 22 minutes, and one quarter of stories are reproduced in under four minutes.
- **Copying fact.** Originality is low: the average article has 36.5% originality and the median is 14.5%.
- **Incentive result.** Originality receives an audience premium: a 50 percentage point increase in originality implies roughly a 40% increase in Facebook shares.

2 Incremental contribution of the paper

- **Universe-level micro data.** The paper builds a transmedia dataset covering virtually all major French online news media.
- **Event-level timeline reconstruction.** It clusters articles into events, identifies the breaking outlet, and measures second-by-second propagation.
- **Measurement of copying.** It traces verbatim copied text to earlier articles, distinguishing original content, internal reuse, external copying, and agency copying.
- **Demand-side integration.** It links production measures to social media shares and audience data to estimate the return to originality.
- **Policy contribution.** It quantifies the appropriability problem in online news and shows that reputation and originality preferences partly mitigate it.

3 Data and measurement

The dataset covers 86 general information media outlets in France and roughly 2.5 million documents published online in 2013. The main analysis uses 851,864 articles classified into 25,215 events. Each event is an event-based story; articles are ordered by publication time to identify the news breaker and reaction time.

3.1 Originality and copying

For each article, the plagiarism-detection algorithm compares text to earlier articles in the same event. The originality rate is

$$Originality_a = \frac{\text{original characters}_a}{\text{total characters}_a}.$$

Interpretation: Originality is measured relative to earlier articles in the same event. The measure is conservative because it captures verbatim copying but not paraphrasing.

3.2 Audience measures

The paper uses article-level Facebook shares and Twitter shares, daily media-level audience data, and a Le Monde dataset that maps article-level shares into article-level views.

Interpretation: This lets the paper move from production-side originality to demand-side returns to originality.

4 Models and empirical specifications

4.1 Article-level returns to originality

$$\log(Audience_{aedn}) = \alpha + Z_{aedn}\beta + \lambda_e + \gamma_n + \delta_d + \varepsilon_{aedn}.$$

Here a indexes articles, e events, d dates, and n media outlets. Z includes publication rank, reaction time, originality, length, original content, non-original content, and news-breaker status.

Interpretation: The identifying comparison is within event, media outlet, and date.

4.2 Mapping shares to views

$$ShareVisits_{adn} = \delta + \sum_{k=1}^6 \gamma_k ShareFacebook_{adn}^k + \eta_{adn}.$$

Interpretation: The Le Monde data are used to construct predicted article-level views for the broader sample.

5 Identification and empirical design

- The production-side results are a measurement exercise: they establish how quickly stories spread and how much text is copied.
- The demand-side results use event, outlet, and date fixed effects to compare articles covering the same story on the same date within the same outlet.
- The main threat is that originality may proxy for unobserved article quality, reporting effort, or photos/graphics bundled with the original article.
- Mechanism tests use high-competition outlets, highly copied outlets, and topic heterogeneity to interpret the originality premium.
- The estimates should be interpreted as returns to original higher-effort content, not as a randomized causal effect of a text-ratio variable alone.

6 Tables and figures

6.1 Table 1: Summary statistics: articles; Table 2: Summary statistics: media outlets

Read: Table 1 reports article length, original and non-original characters, originality, reactivity, and social media shares. Table 2 reports daily and annual media-outlet audience and production measures.

Economic message: The tables show both the scale of the dataset and the skewness of originality and audience.

TABLE 1
Summary statistics: articles (classified in events)

	Mean	Median	SD	Min	Max
Content					
Length (number of characters)	2,467	2,192	1,577	100	98,340
Original content (number of characters)	805	253	1,287	1	53,424
Non-original content (number of characters)	1,661	1,326	1,539	0	48,374
Originality (%)	36.5	14.5	39.8	0	100
Reactivity in hours	41.7	19.1	65.2	0	6,257
Audience					
Number of shares on Facebook	64	0	956	0	240,450
Number of shares on Facebook (winsorized)	37	0	136	0	1,017
Number of shares on Twitter	9	0	42	0	11,908
Number of shares on Twitter (winsorized)	7	0	19	0	126
Obs	851,864				

Notes: The table gives summary statistics. Year is 2013. Variables are values for the articles classified in events. The observations are at the article level. The "Number of shares on Facebook (winsorized)" variable is the version of the Facebook variable winsorized at the 99th percentile. Similarly, the "Number of shares on Twitter (winsorized)" variable is the version of the Twitter variable winsorized at the 99th percentile. Variables are described in more details in the text.

(a) Table 1: article-level summary statistics.

TABLE 2
Summary statistics: media outlets

	Mean	Median	SD	Min	Max
Online audience (daily)					
Number of unique visitors	248,529	107,856	384,001	3,689	2,031,580
Number of visits	340,506	156,735	543,690	4,650	2,945,172
Number of pages views	1,617,616	647,576	2,956,979	12,203	15,203,842
Audience share	1.66	0.72	2.57	0.02	13.65
Facebook (annual)					
Total number of shares	1,137,580	309,176	2,190,098	1,066	13,459,510
Twitter (annual)					
Total number of direct tweets	138,648	27,188	343,000	0	2,464,651
Total number of indirect tweets	3,627	577	8,792	0	58,507
Content (nb of characters) (annual)					
Total content not classified	32,255,744	14,999,537	114,887,872	419,234	1,065,079,600
Total content classified	19,708,659	11,580,943	23,729,089	1,114	101,246,280
Total original content	6,381,766	3,787,462	7,395,088	1,114	31,799,058
Total non-original content	13,326,893	6,860,454	19,705,976	0	76,923,528
Number of breaking news	115	54	174	0	1,011
Observations	85				

Notes: The table gives summary statistics. Year is 2013. Variables are values for media outlets (excepting the AFP and Reuters). The observations are at the media outlet/day level for the online audience statistics (first four rows) at the media outlet/year level for the total number of Facebook shares and the content data.

(b) Table 2: media-outlet summary statistics.

6.2 Table 3: Reaction time; Figure 1: Average reaction time depending on whether the news breaker is a news agency

Read: Table 3 shows that median reaction time is 22 minutes overall and 11 minutes when a news agency breaks the story. Figure 1 shows faster survival decay for agency-broken stories.

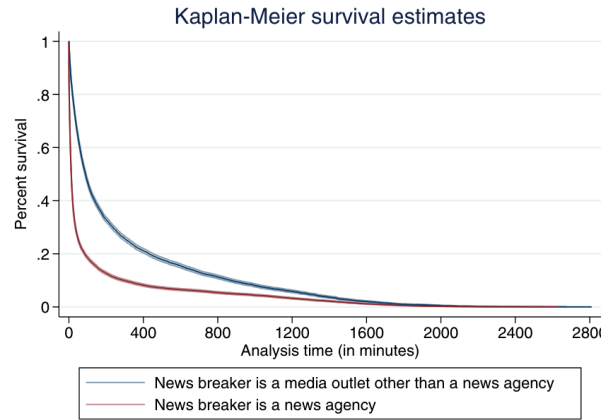
Economic message: News agencies accelerate diffusion and help explain why reactivity is so high.

TABLE 3
Reaction time

	Mean	SD	Median	Min	Max	Obs
Reaction time (in minutes)	169	358	22	0	2,809	25,200
If news breaker is						
Print media	247	400	73	0	2,809	7,200
Television	231	391	57	0	2,098	1,130
Radio	248	398	76	0	2,191	960
Pure online media	394	473	190	0	2,164	510
News agency	116	314	11	0	2,624	15,400

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The tables give summary statistics for the reaction time (in minutes).

(a) Table 3: reaction-time statistics.



(b) Figure 1: Kaplan-Meier reaction-time curves.

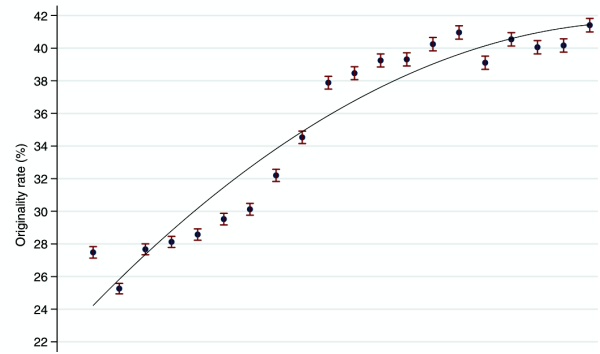
6.3 Figure 2: Originality rate; Figure 3: Correlation between originality and reaction time

Read: Figure 2 shows a bimodal originality distribution. Figure 3 shows that slower articles tend to be more original.

Economic message: Fast reactivity often comes with copying; more original reporting generally takes more time.



(a) Figure 2: originality distribution.



(b) Figure 3: originality by reactivity rank.

6.4 Table 4: Summary statistics: copy

Read: Table 4 decomposes internal copy, external copy, and external copy excluding agencies. Even excluding agencies, 61.8% of event articles contain some external copy.

Economic message: Copying is widespread and can threaten appropriability of news production.

TABLE 4
Summary statistics: copy

	All copy			External copy	Excl. copy from agencies
	(1) Mean/SD	(2) Mean/SD	(3) Mean/SD	(4) Mean/SD	(5) Mean/SD
Originality rate	35.1 (39.0)				
Originality rate wghtd by nb of views (Facebook)		58.0 (38.2)			
Extensive margin					
Share of articles in events with at least some copy			80.0	77.5	61.8
Copying media					
Nb docs copied			4.1 (5.0)	3.9 (4.7)	2.3 (3.3)
External copy rate				61.0 (39.3)	15.9 (25.0)
External copy rate conditional on copying				78.7 (24.5)	25.7 (27.6)
Copied media					
Nb copying docs			3.9 (9.1)	3.3 (8.2)	3.9 (12.5)
% of the doc that is copied				9.4 (17.9)	
% of the doc that is copied conditional on being copied				24.1 (33.9)	
If copied media bk news				53.5 (31.3)	
% of the doc that is copied conditional on being copied					

Notes: The table gives summary statistics. Year is 2013. Variables are values for documents. We consider all the documents classified in events, with the exception of the documents published by the AFP and Reuters. In columns (1) to (3), both internal and external verbatim copying are taken into account. In column (4), we focus on external copy only. In column (5), we focus on external copy and exclude the content copied from the news agencies (the AFP and Reuters). "bk new

Table 4: summary statistics on copying.

6.5 Figure 4: Relationship between unique visitors and Facebook shares; Table 5: Article-level analysis

Read: Figure 4 validates Facebook shares as a proxy for audience. Table 5 shows positive returns to originality for Facebook shares, Twitter shares, and predicted readers.

Economic message: Original producers receive higher audience, mitigating the incentive problem created by copying.

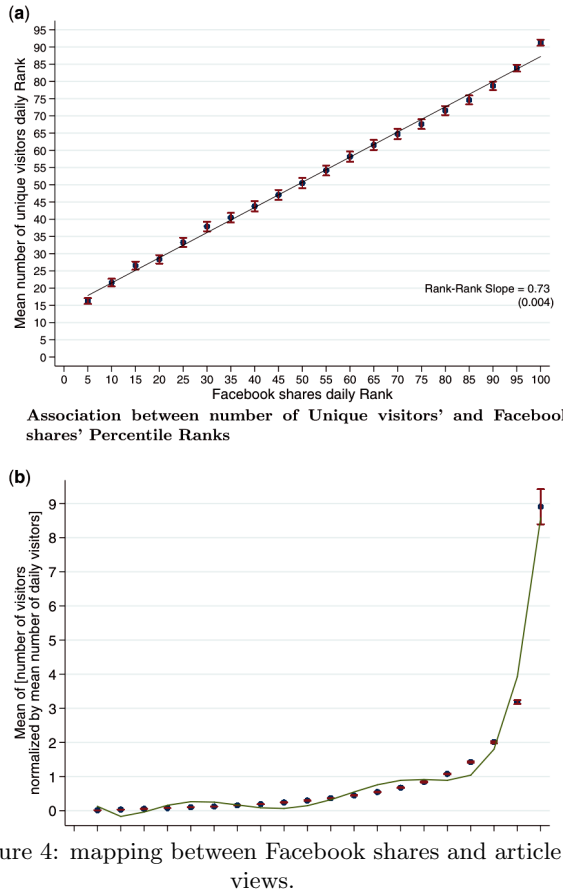


Figure 4: mapping between Facebook shares and article views.

TABLE 5
Article-level analysis: Number of Facebook shares, of Tweets and of predicted readers (log-linear estimations)

	Facebook shares			Number of tweets		Number of predicted readers		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Publication rank	-0.0005*** (0.0001)	-0.0005*** (0.0001)		-0.0002*** (0.0000)	-0.0002*** (0.0000)		-0.0005*** (0.0001)	-0.0005*** (0.0001)
Reaction time (in hours)	-0.0021*** (0.0006)	-0.0021*** (0.0006)		-0.0021*** (0.0001)	-0.0021*** (0.0001)		-0.0045*** (0.0007)	-0.0045*** (0.0007)
Originality rate (%)	0.0068*** (0.0001)	0.0068*** (0.0001)		0.0032*** (0.0001)	0.0032*** (0.0001)		0.0074*** (0.0001)	0.0074*** (0.0001)
Length (thsd ch)	0.0855*** (0.0025)	0.0855*** (0.0025)	0.0808*** (0.0025)	0.0593*** (0.0044)	0.0573*** (0.0044)	0.0796*** (0.0036)	0.0768*** (0.0036)	0.0768*** (0.0036)
Original content (thsd ch)	0.1989*** (0.0033)	0.1989*** (0.0033)	0.1989*** (0.0033)	0.1083*** (0.0010)	0.1083*** (0.0010)	0.2083*** (0.0036)	0.2083*** (0.0036)	0.2083*** (0.0036)
Non-original content (thsd ch)	-0.0286*** (0.0025)	-0.0286*** (0.0025)	-0.0286*** (0.0025)	0.0087*** (0.0014)	0.0087*** (0.0014)	-0.0439*** (0.0032)	-0.0439*** (0.0032)	-0.0439*** (0.0032)
News breaker			0.7684*** (0.0022)			0.4569*** (0.0134)		0.8129*** (0.0232)
Media outlets FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.50	0.50	0.49	0.60	0.60	0.59	0.59	0.49
Adjusted R ²	0.48	0.48	0.47	0.58	0.58	0.47	0.47	0.46
Observations	664,650	664,650	664,650	658,129	658,129	599,378	599,378	599,378
Clusters (event)	25,200	25,200	25,200	25,200	25,200	25,109	25,109	25,109

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the log of the number of times an article is shared on Facebook in Column (1) to (3), the log of the number of times an article is tweeted, retweeted or liked in Column (4) to (6), and the log of the number of views per article in Column (7) to (9). The number of views per article is computed by combining media-level information on the daily number of page views with article-level information on the number of times an article is shared on Facebook (see detailed in Section 5.3). Standard errors in parentheses are clustered by event. Models are estimated using OLS estimations. The unit of observation is an article. All the estimations include media outlets, date, and event fixed effects. Variables are described in more details in the text. "thsd ch" stands for "thousand characters".

Table 5: returns to originality in article-level regressions.

6.6 Figure 5: Facebook shares and originality rate; Table 6: Heterogeneity by competition

Read: Figure 5 shows monotone increases in Facebook shares across originality bins. Table 6 shows that originality has a stronger effect in more competitive environments.

Economic message: The originality premium is stronger when consumers can switch across outlets.

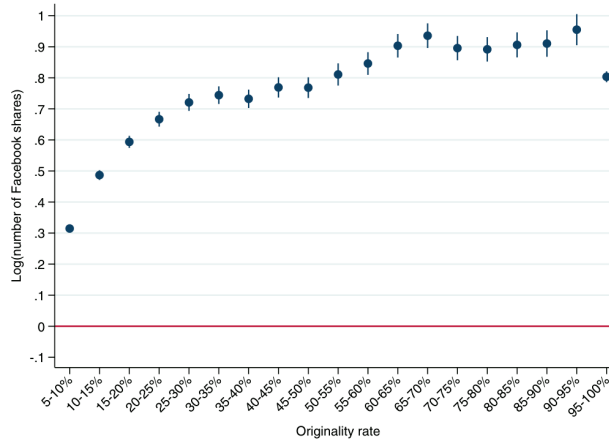


FIGURE 5
Facebook shares and originality rate

es: Figure shows coefficients from a regression of log of the number of times an article is shared on Facebook on twenty categories depending on the originality rate of the articles (articles with an originality rate lower than 5% are the omitted category). Models include media, day, and event fixed effects. Error bars are ± 1.96 standard errors. Standard errors are clustered by event. The t

Figure 5: originality bins and Facebook shares.

TABLE 6
Article-level analysis: number of Facebook shares, of Tweets, and of article views (log-linear estimation), heterogeneity of the effects depending on the competitiveness of the environment

	Facebook shares		Tweets		Number of views	
	(1)	(2)	(3)	(4)	(5)	(6)
Publication rank	-0.0005*** (0.0001)	-0.0004*** (0.0001)	-0.0002*** (0.0000)	-0.0002*** (0.0001)	-0.0004*** (0.0001)	-0.0003 (0.0001)
Publication rank * High competition		(0.0000) -0.0001***		(0.0000) -0.0001***		(0.0001) -0.0001
Reaction time	-0.0135*** (0.0008)	-0.0140*** (0.0008)	-0.0025*** (0.0005)	-0.0025*** (0.0005)	-0.0223*** (0.0011)	-0.0228 (0.0011)
Reaction time * High competition		0.0002 (0.0002)		0.0001 (0.0001)		0.0002 (0.0003)
Originality rate	0.0063*** (0.0001)	0.0052*** (0.0001)	0.0024*** (0.0001)	0.0022*** (0.0001)	0.0072*** (0.0002)	0.0064 (0.0002)
Originality rate * High competition		0.0032*** (0.0002)		0.0006*** (0.0001)		0.0023 (0.0003)
Length	0.1001*** (0.0039)	0.1041*** (0.0057)	0.0736*** (0.0021)	0.0751*** (0.0029)	0.0956*** (0.0055)	0.0827 (0.0075)
Length * High competition		-0.0079 (0.0069)		-0.0040 (0.0033)		0.0307 (0.0095)
Media outlets FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.39	0.39	0.54	0.54	0.26	0.26
Observations	318,196	318,196	310,512	310,512	213,718	213,718
Clusters (event)	24,691	24,691	24,691	24,691	23,846	23,846

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the log of the number of times an article is shared on Facebook in Columns (1) and (2), the log of the number of times an article is shared on Twitter in Columns (3) and (4), and the log of the number of views per article in Columns (5) and (6). The number of views per article is computed by combining media-level information on the daily number of page views with article-level information on the number of times an article is shared on Facebook (as detailed in Section 5.3). Standard errors in parentheses are clustered by event. Models are estimated using OLS estimations. The unit of observation is an article. "High competition" is an indicator variable equal to one for the media outlets that are in a "more competitive" media environment and to zero for those that are in a "less competitive" media environment. The competitiveness of the environment is measured with respect to the

Table 6: heterogeneity by competitive environment.

6.7 Table 7: Heterogeneity by whether the media outlet is copied; Table 8: Heterogeneity by topic

Read: Table 7 shows lower returns to originality for heavily copied outlets. Table 8 shows the originality premium across major topics.

Economic message: Copying weakens appropriability, while the originality premium is robust across topic categories.

6.8 Figure 6: The audience-weighted share of original content

Read: Figure 6 shows that the production share of original content is 32.5%, but audience-weighted originality is much higher, reaching 61.4% under the Facebook-share weighting.

Economic message: Readers disproportionately consume original content, partly compensating original producers.

TABLE 7
Article-level analysis: number of Facebook shares, of Tweets, and of article views (log-linear estimation), heterogeneity of the effects depending on whether the media outlet is copied

	Facebook shares		Tweets		Number of views	
	(1)	(2)	(3)	(4)	(5)	(6)
Publication rank	-0.0005*** (0.0001)	-0.0005*** (0.0001)	-0.0002*** (0.0000)	-0.0002*** (0.0000)	-0.0005*** (0.0001)	-0.0005*** (0.0001)
Publication rank *		-0.0000 (0.0000)		0.0000 (0.0000)		-0.0001* (0.0001)
Highly copied						
Reaction time	-0.0023*** (0.0006)	-0.0019*** (0.0006)	-0.0021*** (0.0003)	-0.0018*** (0.0003)	-0.0045*** (0.0007)	-0.0044*** (0.0007)
Reaction time *		-0.0003* (0.0002)		-0.0003*** (0.0001)		0.0001 (0.0002)
Originality rate	0.0068*** (0.0001)	0.0083*** (0.0001)	0.0032*** (0.0001)	0.0052*** (0.0001)	0.0074*** (0.0001)	0.0087*** (0.0002)
Originality rate *		-0.0022*** (0.0002)		-0.0030*** (0.0001)		-0.0019*** (0.0002)
Highly copied						
Length	0.0855*** (0.0025)	0.0644*** (0.0030)	0.0593*** (0.0014)	0.0505*** (0.0017)	0.0796*** (0.0030)	0.0680*** (0.0041)
Length * Highly copied		0.0404*** (0.0038)		0.0174*** (0.0022)		0.0212*** (0.0049)
Media outlets FE	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.40	0.41	0.54	0.55	0.42	0.42
Observations	664,650	663,825	656,129	655,304	509,378	508,665
Clusters (event)	25,200	25,200	25,200	25,200	25,109	25,109

textitNotes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the log of the number of times an article is shared on Facebook in Columns (1) and (2), the log of the number of times an article is shared on Twitter in Column (3) and (4), and the log of the number of views per article in Columns (5) and (6). The number of views per article is computed by combining media-level information on the daily number of page views with article-level information on the number of times an article is shared on Facebook (as detailed in Section 5.3). Standard errors in parentheses are clustered by event. Models are estimated using OLS estimations. The unit of observation is an article. "Highly copied" is an indicator variable equal to 1 for the media outlets that have been highly copied in 2013, *i.e.*, whose share of the center that has been copied is higher than the median, and to 0 otherwise (see the text for more details). All the estimation include media outlets, date, and event fixed effects. Variables are described in more details in the text. "(thsd ch)" stand for "thousand characters."

Table 7: heterogeneity by being highly copied.

TABLE 8
Article-level analysis: number of Facebook shares (log-linear estimation), heterogeneity of the effects depending on the topic of the events

	All	Crime	Politics	Economy	Arts	Sport	Disaster	Other
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Publication rank	-0.0008*** (0.0001)	-0.0008*** (0.0001)	-0.0008*** (0.0001)	-0.0008*** (0.0002)	-0.0021*** (0.0001)	-0.0010*** (0.0001)	-0.0023*** (0.0004)	-0.0010*** (0.0002)
Reaction time	-0.0023*** (0.0006)	-0.0019*** (0.0006)	-0.0021*** (0.0003)	-0.0018*** (0.0003)	-0.0045*** (0.0007)	-0.0044*** (0.0007)	-0.0019*** (0.0002)	-0.0019*** (0.0002)
Originality rate (%)	0.0068*** (0.0001)	0.0083*** (0.0001)	0.0032*** (0.0001)	0.0052*** (0.0001)	0.0074*** (0.0001)	0.0087*** (0.0002)	0.0068*** (0.0001)	0.0087*** (0.0002)
Length	0.0855*** (0.0025)	0.0644*** (0.0030)	0.0593*** (0.0014)	0.0505*** (0.0017)	0.0796*** (0.0030)	0.0680*** (0.0041)	0.1131*** (0.0081)	0.0920*** (0.0052)
Media outlets FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.50	0.48	0.53	0.48	0.52	0.50	0.49	0.53
Adjusted R ²	0.48	0.47	0.52	0.46	0.49	0.47	0.47	0.51
Observations	664,650	15,658	121,031	121,054	63,795	56,381	39,038	113,693
Clusters (event)	25,200	5,132	1,422	5,093	2,854	2,640	1,995	4,694

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the log of the number of times an article is shared on Facebook. In columns (1), all the events in our sample are included in the estimation. Column (2) reports the estimates for the "Crime, law and justice" events, Column (3) for the "Politics" events, Column (4) for the "Economy, business and finance" events, Column (5) for the "Arts, culture and entertainment" events, Column (6) for the "Sport" events, Column (7) for the "Disaster and accident" events, and Column (8) for the events classified in all the other IPTC categories. The topics correspond to the IPTC media topics described in the article and defined in the Supplementary Appendix Section C5. Standard errors in parentheses are clustered by event. Models are estimated using OLS estimations. The unit of observation is an article. All the estimations include media outlets, date, and event fixed effects. Variables are described in more details in the text. "(thsd ch)" stands for "thousand characters."

Table 8: heterogeneity by topic.

FIGURE 6
The audience-weighted share of original content

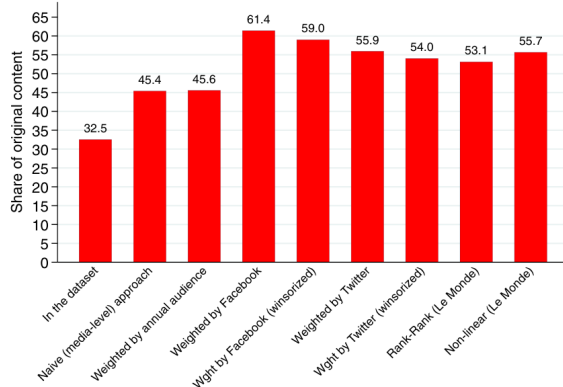


FIGURE 6

The audience-weighted share of original content

Figure 6: The figure reports the audience-weighted share of original content we obtain using our different approaches to compute the audience-weighted share of original content. The first bar ("In the dataset") simply reports the share of original content in the dataset (with no weight). "Naive (media-level) approach" reports the share of original content we obtain when we attribute the same number of views to each article. The third bar ("Weighted by annual audience") reports the share of original content when we weight the content of each article by the average annual audience of the media outlet in which it was published. The fourth bar ("Weighted by Facebook"), we attribute number of views to articles assuming a linear relationship between the number of Facebook shares and the number of article views. The fifth bar ("Wght by Facebook (winsorized)") relies on the same methodology but with the winsorized version of the Facebook shares variable. The sixth ("Weighted by Twitter") and seventh ("winsorized") bars are computed similarly than the fourth and fifth bars, except that we use the number of shares on Twitter instead of Facebook.

Figure 6: audience-weighted original content.

7 Short summary

- The paper studies online news production and copying in France in 2013.
- It constructs event timelines and measures verbatim copying at scale.
- Online news is very reactive but heavily copied.
- Originality receives a positive audience premium.
- This premium is stronger in competitive environments and weaker for heavily copied outlets.
- Audience-weighted originality exceeds production-weighted originality.

8 Conclusion

8.1 Core conclusion

Online news markets feature massive copying, but original producers receive higher audience, so the market contains a partial incentive-restoring mechanism.

8.2 Incremental contribution revisited

The paper’s incremental contribution is to jointly measure information production, copying, propagation, and consumption in a single event-level dataset.

8.3 Policy implications

The results matter for copyright and neighboring-rights policy because they show both a real appropriability problem and a demand-side reward to originality.

8.4 Limitations

The measures capture verbatim copying, not paraphrasing; social media shares are imperfect audience proxies; and the setting is France in 2013.

8.5 Takeaway

The digital news market is an intermediate equilibrium: copying is fast and extensive, but originality still matters for demand.